

Enhancing Cultural Understanding on Chinese Social Media: A RAG-based Approach with CoT Reasoning Analyzing Absurd Language Style

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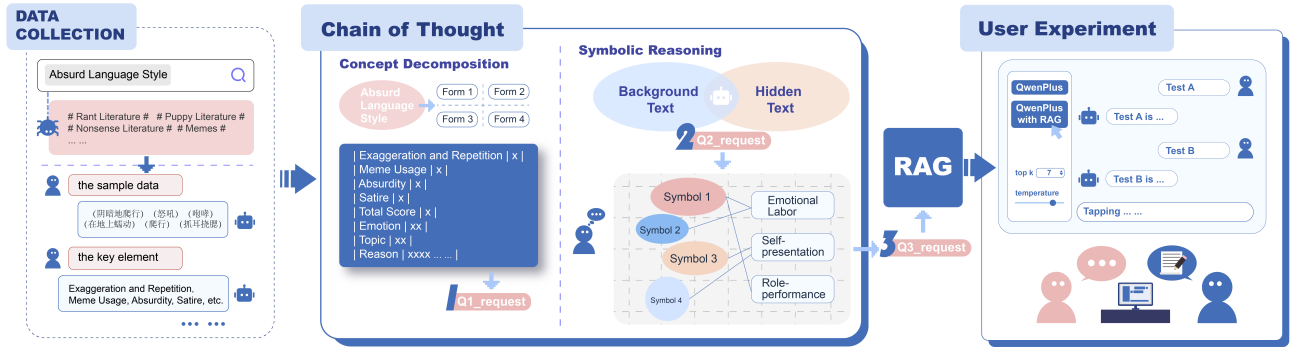


Figure 1: A dialog logic chart explains the entire process, including how the Chain-of-Thought (CoT) and Retrieval-Augmented Generation (RAG) methods work together. CoT results are fed into RAG for knowledge retrieval, enabling the development of a dialogue system that supports cultural understanding. The effectiveness of the cultural understanding framework is ultimately validated through user experiments.

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Abstract

The absurd language style on Chinese social media is a unique cultural phenomenon that highlights the importance of cultural understanding in digital communication. This study introduces a Retrieval-Augmented Generation (RAG)-based conversational system that applies concept decomposition and symbolic reasoning through the Chain-of-Thought (CoT) method. A database of 120,000 text samples was built to support the system, designed to help Sino-digital non-natives better understand absurd language on social media. Experiments show that the RAG-based system surpasses

traditional LLMs in comprehension, usability, and accuracy. Semi-structured interviews further evaluated the system from three aspects: user behavior, cultural needs, and system performance. This study offers a practical approach to analyzing viral texts within specific cultural contexts and proposes a reusable system for cultural understanding, providing new directions for cross-cultural communication and digital literacy.

CCS Concepts

• **Human-centered computing** → **Collaborative and social computing devices**; **Empirical studies in collaborative and social computing**; **Empirical studies in HCI**.

Keywords

Chain of Thought, Retrieval-Augmented Generation, Symbolic reasoning, Large Language Model

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1 Introduction

The digital age has transformed human communication, generating new online language phenomena that function as dynamic symbolic systems evolving through social interactions and cultural contexts [1]. Symbolic interactionism provides a framework for understanding how these digital linguistic styles create social and cultural significance [7]. These phenomena spread online through shared humor, satire, and intertextuality [22, 36], with social media platforms like WeChat, Weibo, Twitter, and Reddit facilitating rapid linguistic evolution [35, 47] shaped by distinct cultural attributes [17, 23, 24, 29, 44]. A notable example is "absurd language" on Chinese platforms like Weibo, characterized by irony and nonsensical humor [10], which emerged during COVID-19 as a collective commentary style paralleling Western "memes" [6, 22]. Through this medium, users engage in symbolic resistance and self-expression, reflecting Goffman's 'role performance' [20]. While memes transcend linguistic barriers, absurd language operates within China's collectivist cultural framework, which promotes shared references [15] through vernacular dialects and cultural humor [33], both generating 'conviviality' through shared practices [6].

Digital tools play a vital role in acculturation—a dual process of cultural and psychological adaptation—by enabling communication and cultural engagement on a global scale [5, 43, 46]. The "TikTok refugees" phenomenon, for example, and the rapid launch of AI translation features by China's social media platform Xiaohongshu represent a typical case of cross-cultural communication on the internet. The "TikTok refugees" phenomenon specifically highlights the potential of digital platforms to foster mutual understanding and cultural integration, demonstrating how contemporary cultural exchange is deepening and overcoming barriers. [39]. However,

marginalized groups often face significant challenges in engaging on these platforms. Older adults, for instance, can struggle with generational and cognitive barriers, while "digital immigrants" often lack familiarity with the social-media-specific language and cultural practices required for effective communication, creating a 'second-level digital divide' [21, 41]. For "Sino-digital non-natives" - individuals struggling with culturally specific digital practices, including expats such as international students, older users, and digital immigrants - the absence of accessible, inclusive tools therefore risks perpetuating digital and social exclusion in an increasingly interconnected world.

This study explores challenges faced by Sino-digital non-natives and Digital immigrants in understanding absurd language on Chinese social media. We utilize large language models (LLMs) to analyze absurd language, proposing a framework to address its cultural and emotional significance. We integrate the logic of concept breakdown and symbolic reasoning into the Chain-of-Thought (CoT) method to structurally analyze absurd language texts crawled from the web. Results are stored in a database and retrieved using Retrieval-Augmented Generation (RAG), ultimately forming a preliminary dialogue system for cultural understanding. By combining computational efficiency and cultural sensitivity, this approach extends beyond conventional sentiment analysis to explore the role of absurd language in cross-cultural engagement and inclusion [8, 41]. The framework acknowledges the importance of providing users with tools tailored to their unique challenges, emphasizing practical applications for digital literacy, social integration, and intercultural understanding.

Our contributions are as follows:

- Apply a concept decomposition and symbolic reasoning method combining Chain-of-Thought based logic to help LLMs understand contemporary Chinese cultural phenomena. Establish an absurd language database with 120,000 analyzed texts.
- Develop a dialogue system to establish culture understanding, using Retrieval-Augmented Generation to support conversations.
- Collect user feedback to inspire further tool refinement through qualitative and quantitative user research.

2 Related Work

2.1 LLMs in Cultural Research and Social Media Analysis

Research on the applications of large language models in cultural studies and social media has demonstrated the ability of LLMs, such as GPT-4 and mBERT, to model linguistic phenomena and address cultural differences. For example, Liu et al. introduced an approach that allows LLMs to refine semantic annotations and expand their knowledge base gradually, enabling them to adapt to the evolving nature of social media texts. The incremental incorporation of new data and contexts allows for LLMs to handle challenges of informal communication, which is essential for cross-cultural understanding [26]. Furthermore, Li et al.'s application for reducing cultural bias in LLMs aims to promote cultural understanding through multi-agent simulations by generating cultural datasets for education and moderation [25]. Similarly, Toxtli et al. leveraged

theories like Chronemics to introduce a culturally-aware AI tool for crowdworkers, which takes different cultural work systems into account [40]. However, these studies focus on structured environments rather than informal communication styles as they are found in social media interactions.

2.2 Symbolic Reasoning and CoT Prompting in Social Media Contexts

By integrating symbolic reasoning, LLMs improve tasks such as sarcasm detection, metaphor interpretation, and sentiment analysis [9]. In social media contexts, Gatto et al. proposed a CoT embedding method for stance detection [19], while Weinzierl and Harabagiu combined CoT prompting with In-Context Active Curriculum Learning to analyze communication frames [45]. Vishwamitra et al. used CoT reasoning to tackle online hate on platforms like X (formerly Twitter) [42].

However, these methods mainly target structured domains such as commonsense reasoning and semantic role inference. Cultural nuances, idiomatic expressions, and code-mixed language complicate LLM performance on non-standardized and context-dependent data [13], as models not only need to recognize such content, but also contextualize it within its broader cultural frameworks to address phenomena such as absurd language.

2.3 RAG Integration and Knowledge Bases in Cultural and Social Media Contexts

RAG methods, commonly used for question-answering tasks, improve LLM capabilities by integrating parameterized knowledge with extensive non-parameterized external data [18]. RAG-based chatbots have been developed for enterprises [3], education [4, 28], and healthcare [34]. In the context of cross-cultural applications, Conia et al. used multilingual knowledge graphs for nuanced translations [14]. Pichai applied a modular RAG framework for cultural preservation [31], while Chang et al. merged LLMs with RAG to develop a Hakka-specific chatbot using cultural knowledge bases [11]. In social media, Cheng et al. predicted social media popularity with an aspect-aware retrieval hypergraph [12], while Ahn et al. applied RAG to analyze informal and context-rich internet conversations, such as those on Reddit threads [2].

These studies demonstrate the effectiveness of RAG in the context of cultural understanding and social media language analysis. However, research on informal, evolving styles like Chinese absurd language remains limited, with existing frameworks often overlooking the emotional and symbolic dimensions crucial for social media analysis [7, 13].

3 Methodology

The dataset of 120,000 text segments was collected from Chinese social media platforms (Weibo and Xiaohongshu) using the Scrapy framework, filtered by tags associated with absurd language styles (e.g., crazy literature). To maintain cultural consistency, we focused on content relevance rather than bloggers' demographic backgrounds. A later analysis of blogger profiles (e.g., language patterns, location tags, and interaction communities) suggested that most content was generated by bloggers deeply engaged with Chinese internet culture [6]. This approach aligns with our goal of

analyzing culturally specific phenomena in China's digital ecosystem.

Based on these texts, we formulated structured questions using the CoT format to adopt a cultural research perspective. This method encourages the model to reason step by step when processing each text. We followed by using the API of Qwen-Plus for text generation. The Qwen series of models have demonstrated remarkable performance across multiple benchmark tests, showing particularly strong capabilities in complex reasoning and contextual understanding tasks [32]. The generated responses were then stored in an offline knowledge base.

For our experiment, we adopted a zero-shot approach, allowing the model to generate outputs without task-specific training data. Prompts were constructed in a template format. The response to each question was incorporated into the input for the next question, allowing the model to generate content progressively while maintaining contextual coherence. Our logic of how CoT and RAG methods work together is shown in Figure 2.

3.1 Chain-of-Thought Prompting Process

During the CoT prompting process, we first introduced a method of concept decomposition and quantification. This approach breaks down and quantifies the features of a concept to assist the LLM in understanding abstract ideas. Questions designed for the LLM addressed both surface-level perspectives (text representation) and deeper-level perspectives (text connotation).

Second, we incorporated a method of symbolic reasoning. This method helps the LLM identify specific symbols in the text's rhetoric that carry notable emotional characteristics and, ultimately, interpret the psychological needs behind the actions of the text's creator. The steps of our CoT process are as follows:

- **Step 1:** Approaching from a textual angle, we segmented the text and extracted keywords, breaking the theme of "absurdity" into four typical features: exaggeration and repetition, satire and sarcasm, absurdity, and meme references. The LLM was instructed to score each feature individually, and the overall score was used to map the degree of absurdity in the text. The results were stored as Q1_request.
- **Step 2:** Expanding to a contextual dimension, Q1_request was used as input to prompt the LLM to identify potential background information (e.g., social, cultural, historical) and determine if implicit content existed. An explicit instruction was given: the higher the total score in Q1_request, the more likely it is that implicit content exists. The final results were stored as Q2_request.
- **Step 3:** Using Q1_request and Q2_request as inputs, we introduced the knowledge framework of creative labor theory to explore the psychological motivations behind creative behaviors [27, 30, 37, 38]. The LLM was prompted to analyze the relationships among textual forms, emotional symbols, and psychological motivations, such as role-playing, emotional labor, and self-presentation, to establish a behavior analysis model.

3.2 Retrieval-Augmented Generation (RAG) Embedding Process

We utilized the bge-large-zh-v1.5 model to generate sentence embeddings for each post and stored these high-dimensional vectors

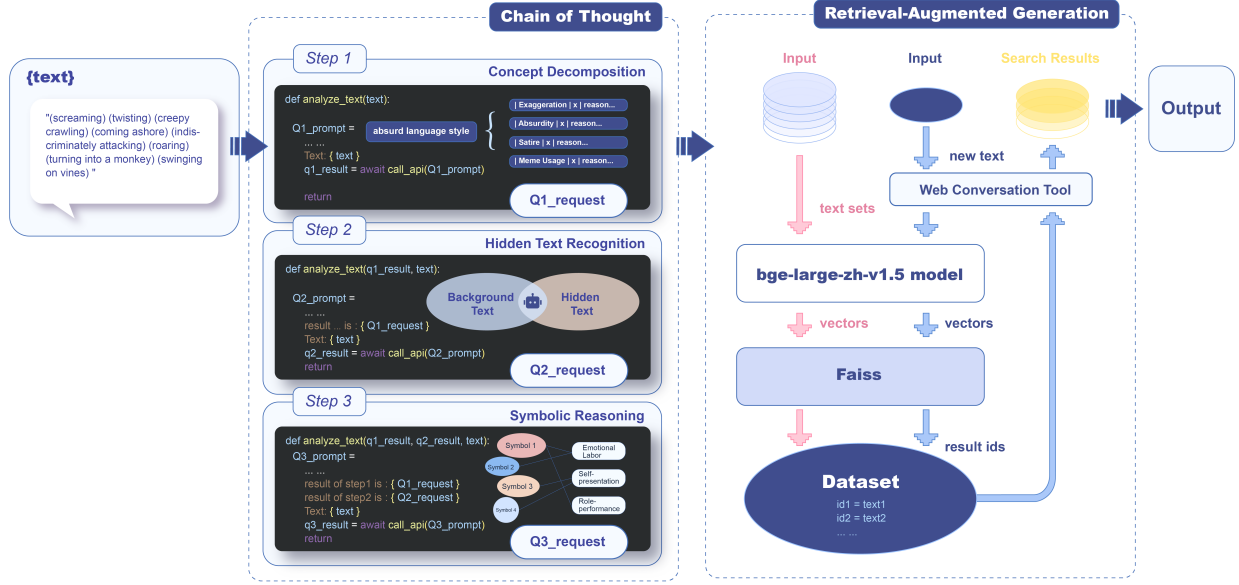


Figure 2: A dialog logic chart explains how the CoT and RAG methods work together.

in a Faiss database for efficient retrieval. Each sentence embedding was mapped one-to-one with its corresponding CoT response. Only the original data in the vector database is stored to ensure that the retrieval process focuses on identifying original data similar to the new input, as included responses might reduce recall accuracy, and LLM-generated responses often exceed the maximum token limit of the embedding model due to their length.

During user input, the text to be analyzed is used as a query vector to perform similarity search within the vector database. We rank the retrieved texts using a weighted score combining vector similarity and the BM25 algorithm. The top three most relevant texts are selected as examples, incorporated into a prompt template, and then input into the LLM.

4 User Experiment

We conducted a preliminary evaluation of our dialogue system through user research to assess its effectiveness in helping participants understand absurd language and to gather constructive feedback, which would guide directions and principles for further development. The following sections detail the participants' characteristics, experimental procedures, evaluation methods, and corresponding findings.

4.1 Participants

To ensure sample diversity, we recruited 6 participants through both online and offline channels, taking into account the emergence timeline of absurdist literature and user age distribution. This resulted in a participant group spanning various age groups and cultural backgrounds. Such diversity helped us gain comprehensive insights into user experiences and needs across different demographics, laying a foundation for future iterations. Each participant engaged in a 60-minute experimental session and received compensation of 60

RMB (approximately 8.5 USD). The detailed characteristics of the participants are presented in the table below.

4.2 Procedure

Each participant's experimental session lasted approximately 60 minutes, structured as follows:

- **Introduction (5 minutes):** Researchers provided a brief overview to ensure participants fully understood the experimental objectives, procedures, and guidelines. During this phase, participants also completed a basic demographic questionnaire.
- **Comprehension Test (30-40 minutes):** The core phase consisted of six absurdist literature text comprehension tasks. Participants engaged in a dialogue with the system to understand the target texts until reaching their self-perceived comprehension saturation point or acknowledging comprehension barriers. The six absurdist literature texts were selected based on two criteria: (1) salient absurdity features derived from our CoT-based concept decomposition (Section 3.1), and (2) high comprehension complexity identified through expert evaluation. These texts reflect culturally specific phenomena on Chinese social networks, as their humor relies on shared knowledge of Chinese internet subcultures (e.g., homophonic puns in Mandarin [33]). This design intentionally isolates cultural specificity to assess the system's ability to decode absurd language rooted locally. To evaluate system effectiveness, we employ an AB testing methodology, where participants interacted with both a conventional LLM (Qwen-plus) and an enhanced system based on Retrieval-Augmented Generation, with texts randomly assigned. After completing each text comprehension task, participants immediately completed an evaluation scale that

Participant	Gender	Age	Understanding Level(1-5)	Major	Nationality
P1	Male	24	3	Computer Science	China
P2	Female	21	5	Computer Science	China
P3	Male	26	2	Liberal Arts	Barbados
P4	Female	24	5	Psychology	USA
P5	Female	27	3	Philosophy	China
P6	Male	43	3	HCI	China

Table 1: Participant Demographics and Characteristics

assessed their understanding on multiple dimensions, including text meaning comprehension, application scenario awareness, and mastery level.

- **Semi-structured Interview (20 minutes):** Based on the observed dialogue process and the details of the interaction, the researchers conducted in-depth discussions about user experience, system functionality, and improvement suggestions to collect qualitative feedback.

Throughout the session, we observed and documented the dialogue progress and interaction rhythm, providing substantial evidence for subsequent analysis. The collected scale data and interview content would serve as the foundation for system optimization and further iterations.

4.3 Result

Through quantitative analysis of the scales, we found that Group B, utilizing RAG technology, outperformed Group A (traditional LLM) across three dimensions (see Figure 3). In terms of user comprehension, Group B achieved a higher mean score of 4.30 (SD = 0.67) compared to Group A's 3.90 (SD = 0.89). Regarding system proficiency, Group B scored 4.00 (SD = 0.71), surpassing Group A's 3.60 (SD = 0.84). For AI comprehension capability, Group B demonstrated superior performance with a mean score of 3.10 (SD = 1.09) versus Group A's 2.50 (SD = 1.18). Regarding the accuracy of participants' understanding, Post-system interaction accuracy rates further supported these findings, with Group B achieving 96.67% accuracy (90%, 100%, 100%) compared to Group A's 78.33% (85%, 70%, 80%). These results indicate that our system effectively facilitated participants' transition from non-comprehension to comprehension of test phrases.

We systematically coded semi-structured interview results to explore user behaviors, needs, and experiences when using the dialogue system to learn and understand absurd language style. We analyzed interview content line by line to label concepts related to system usage, and finally grouped related concepts into three main dimensions: user behavior, conversational needs, and system performance.

To ensure measurement validity, we combined Post-system interaction accuracy rates with expert validation. Independent experts recorded behavioral cues (e.g., query patterns) during interactions and cross-checked interview responses against pre-annotated cultural meanings. Finally, participants confirmed the expert-validated interpretations. This triangulation method [16] strengthened the reliability of both system performance and user understanding metrics.

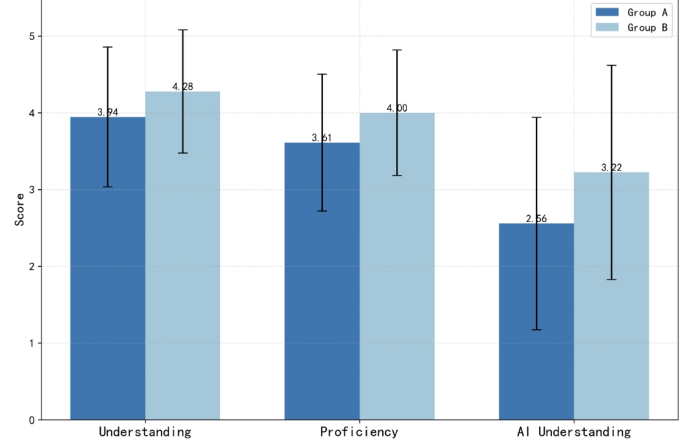


Figure 3: Comparison of Understanding Level, Usage Proficiency, and AI Comprehension Level.

- **User behavior:** Includes inquiry and search activities. Inquiry behavior mainly involves exploratory, comparative, and creative questioning. Exploratory questions rely on divergent thinking based on system-provided text, while comparative and creative questions prompt the system to mimic styles or generate new themes. Some participants exhibited "conversation blind spots" (P3: "I don't know what to ask first, so I start by asking the AI's thoughts"). Search behavior refers to finding solutions for unfamiliar cultures. Participants often start by searching social platforms for others' interpretations (P5: "I check comments or search similar posts using keywords") and then use external systems or ask friends for further clarification (P3: "I ask my younger Chinese friends about unfamiliar memes").
- **Conversational Needs:** Includes the need for cultural understanding, meme updates, and entertaining interactions. Participants found system-provided texts helpful (P2: "It helps me quickly understand absurd language styles"; P4: "It gave me new ideas") and logically structured (P2: "I think about why people write this way and their real intentions, which matches CoT reasoning"). The system can assist with trending phrases but is not always accurate (P5: "AI struggles with homophonic puns"). It can mimic absurd styles but lacks genuine absurdity, often producing rigid responses (P6: "It imitates form but struggles with creative text structures, like unique language order and structure").

- **System performance:** The system's performance is evaluated based on knowledge completeness, content accuracy, and emotional adaptability. Participants noted that the system provides comprehensive knowledge, and prior texts do not restrict questioning (P1: "It offers complete background knowledge"; P2: "I use it as a reference"). However, the content is not entirely accurate (P1: "It can grasp the true emotions of the text but struggles to create similar absurd styles"; P4: "Misunderstanding of memes can lead to incorrect sentence understanding"). Additionally, the system lacks emotional adaptability (P6: "Its responses should reflect absurd personality traits"; P5: "I expect it to be more crazy").

5 Discussion

5.1 System Performance and Improvements

The results of our user experiments showed that, compared to traditional LLMs, the system effectively helped users understand absurd language, especially by explaining the hidden meanings behind internet memes and linguistic styles. This was particularly beneficial for non-native Chinese speakers. The system's structured approach, which analyzed text across linguistic dimensions and symbolic meanings, aided in the comprehension of long absurd texts. The system also provided quick feedback with usage scenarios and examples, which most participants found essential for understanding this linguistic style.

However, the system also has some notable issues. First, the texts generated by the LLM using the CoT method tend to be burdensome to read, often lengthy and overly academic, making them difficult for users to quickly process. Second, the LLM has not yet fully captured the deeper meanings of absurd language, a limitation that is particularly evident in tasks involving imitation. Additionally, the system lacks interactive engagement, leaving significant room for improvement in user experience and emotional interaction.

To improve the system's efficiency, optimizations can be made in several areas. First, the CoT prompt design can be refined to reduce iterations and decrease the cognitive load of the output text. This may involve adopting more concise prompt templates or generating more streamlined reasoning steps through pre-trained models, thereby shortening user interaction time. Second, redesigning user interactions is essential. One potential approach is to embed the system as a plugin within commonly used social media or learning tools, allowing users to access features exiting their current applications. Additionally, the system should provide a more engaging and efficient experience, such as guided questioning, relevant blog examples, and the use of images or memes to explain scenarios. A more interactive experience for users could be achieved through language style imitation, and incorporating personality traits could strengthen emotional resonance, leading to an overall improved interaction.

5.2 Social Impact and Application Potential

"Absurd Language," as a cultural phenomenon in Chinese online spaces, uniquely combines humor, absurdity, and social critique. Understanding this style is both a linguistic and cultural challenge, as it reflects the collective emotions, frustrations, and creativity of internet users. By decoding and engaging with this style, our

research bridges the gap between traditional linguistic analysis and contemporary online communication.

This work promotes cross-cultural understanding and digital literacy by making absurd language more accessible, reducing misunderstandings, and fostering inclusive online interactions between Sino-digital natives and non-natives. Beyond individual users, the system can be integrated into educational platforms to teach digital literacy and cultural awareness, assist content creators in generating culturally relevant material, and analyze other forms of online communication, such as memes, internet slang, and political satire. These applications offer valuable insights into the evolution of digital discourse.

6 Conclusion

This study demonstrates the effectiveness of a Retrieval-Augmented Generation (RAG)-based system with Chain-of-Thought (CoT) reasoning to assist in the understanding absurd language styles on Chinese social media. By integrating concept decomposition, symbolic reasoning, and retrieval-enhanced workflows, the system enables Sino-digital non-natives to better comprehend cultural nuances and linguistic phenomena. Experimental results showed that the RAG-based system surpassed traditional LLMs in user comprehension (4.30 vs. 3.90), system proficiency (4.00 vs. 3.60), and AI comprehension capability (3.10 vs. 2.50). Post-interaction accuracy rates highlighted its effectiveness, with 96.67% accuracy compared to 78.33% for traditional LLMs. Further insights highlighted that the system excels in explaining the implicit meanings and cultural contexts of absurd language, with significant benefits for non-native speakers. While the system exhibits limitations in emotional adaptability and text brevity, it lays a promising groundwork for subsequent refinements such as interactivity, emotional resonance, and integration into widely used platforms.

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